Linear Regression on US Housing Price

What can be determined from this plot?

The predicted prices lie close to the red diagonal line (predicted = actual). It forms a tight, roughly straight band. This means the model's predictions track the true house prices well. This agrees with the high R^2 of about 0.92 on the test set.

The spread around the line is fairly uniform across the whole price range, with no obvious curve or funnel shape, so the model does not over- or under-predict at low or high prices.

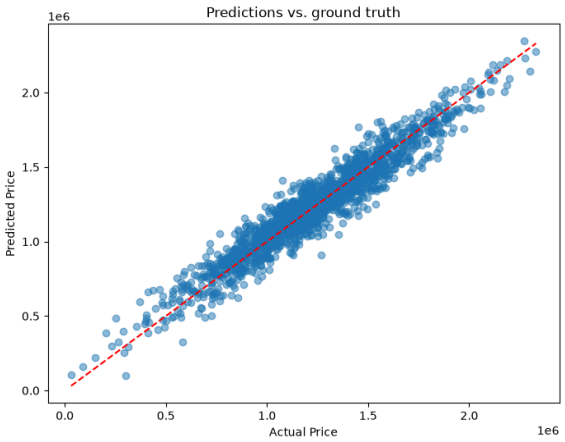


Figure 1. Plot predictions against the ground truth values

Ridge/LASSO polynomial regression with linear and random sampling

Linear vs. random sampling comparison

Random sampling scores higher than linear sampling at every polynomial degree. The gap is largest at low degrees. At degree 2, linear sampling's R^2 is negative, while random's is positive. From degree 4 onward, both curves level off above 0.8 and the gap narrows. The random-sampling curve is also smoother, zigzags less, and settles near 0.98. Randomly placed points may cover the function's shape better than evenly spaced ones on this small dataset.

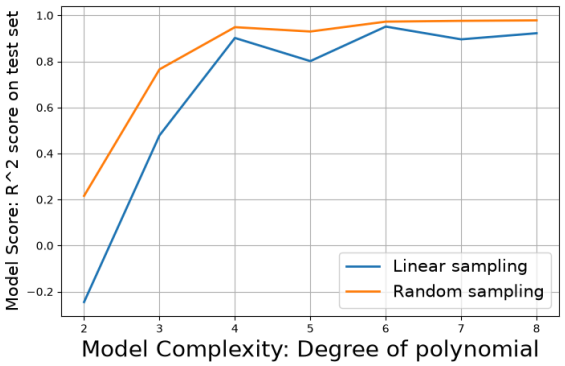


Figure 2. Linear and random sampling comparison

Support Vector Machine practice notebook with breast cancer data set

Comparison between the optimized model performance vs. the default SVC

The tuned model (C=1, gamma=0.0001) improves modestly on the default SVC. Accuracy rises from 0.92 to 0.94, and errors fall from 13 to 11 of 171 test cases. It correctly classifies more benign cases (59 vs. 56 of 66), so benign recall rises from 0.85 to 0.89. Malignant recall is nearly unchanged (0.96 vs. 0.97). Precision and F1 improve slightly for both classes. The default model's baseline was already strong, so tuning gave only a small gain.

	precision	recall	f1-score	support
0	0.94	0.89	0.91	66
1	0.94	0.96	0.95	105
accuracy			0.94	171
macro avg	0.94	0.93	0.93	171
weighted avg	0.94	0.94	0.94	171

Figure 3. Classification report